Power and resource tradeoffs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Power and resource tradeoffs

Distinguish supplied alternative-trial frequencies, per-study cost and unmeasured null error.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **power** and resource tradeoffs

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of pilot experiments

Reference example, not the only acceptable wording: Teacher explanation: A pilot checks feasibility, measurement and failure handling before the primary study. Pilot results used for revision are development **evidence**.

Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of privacy and data governance

Reference example, not the only acceptable wording: Teacher explanation: Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sensitivity analysis

Reference example, not the only acceptable wording: Teacher explanation: Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices.

It should not secretly select the most favourable result. Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range.

Settings affecting data roles require special caution.

Pictured end goal

Explain **power** and resource tradeoffs using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Pilot experiments

Privacy and data governance

Sensitivity analysis

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Power** concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **power** and resource tradeoffs

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Planning item	Complete main example
Alternative model	Fixed supplied alternative and test; model details not provided for reproduction
Trial counts	One hundred alternative simulations per design, stipulated
Small design	n = 20 independent units; 62 detections
Large design	n = 40 independent units; 83 detections
Unit cost	Three credits per independent unit
Per-study cost	n × 3, not the total cost of all one hundred simulation trials

Planning item	Complete main example
Unknown quantities	Null false-positive rate, simulation uncertainty interval, real-study power and actual compute cost
Separate picture	Different supplied counts: 60/100 and 80/100, unit cost two resource units

Main evidence ledger and checked key

Main quantity	n = 20	n = 40
Supplied detections / alternative trials	62/100	83/100
Estimated detection frequency	62%	83%
Independent units per study	20	40
Per-unit cost	3 credits	3 credits
Per-study cost	20 × 3 = 60 credits	40 × 3 = 120 credits
Large minus small	21 percentage points more observed detection	60 extra credits per study
Null error and actual-study guarantee	Not supplied	Not supplied

Main: 62% versus 83% supplied alternative-trial detection frequencies, a twenty-one-percentage-point difference. Per-study costs are sixty and one hundred twenty credits, an extra sixty. Picture: 60% versus 80%, cost forty versus eighty resource units, twenty points for forty extra units. These are different supplied scenarios. Changed effect or variability needs new planning; no replacement frequency is known.

Independent task: 70/100 versus 85/100 gives 70% versus 85%, fifteen points; n = 30 or 60 at two per unit costs sixty or one hundred twenty. Null error is unmeasured, finite-trial estimates are noisy, and no real-study guarantee or universal sample-size claim follows.

Power and resource tradeoffs — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits.	Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.	Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured. Qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits.

Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Power/budget probe	Checked interpretation
62 detections out of 100 alternative trials	62% estimate, not known exact power
No null trials	False-positive frequency is unmeasured
Per-study cost versus all simulation work	Different denominators; do not confuse them
More units under fixed assumptions	Compare supplied frequencies and cost; no universal rule
Effect or variability changes	Recompute under the new model; old power need not transfer
Different unit IDs	Does not itself establish independent sampling
Higher observed detection rate	Not a guarantee the next study succeeds
Simulation model unspecified	Do not pretend supplied counts can reproduce the simulation

Separate pictured facts and full reasoning

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

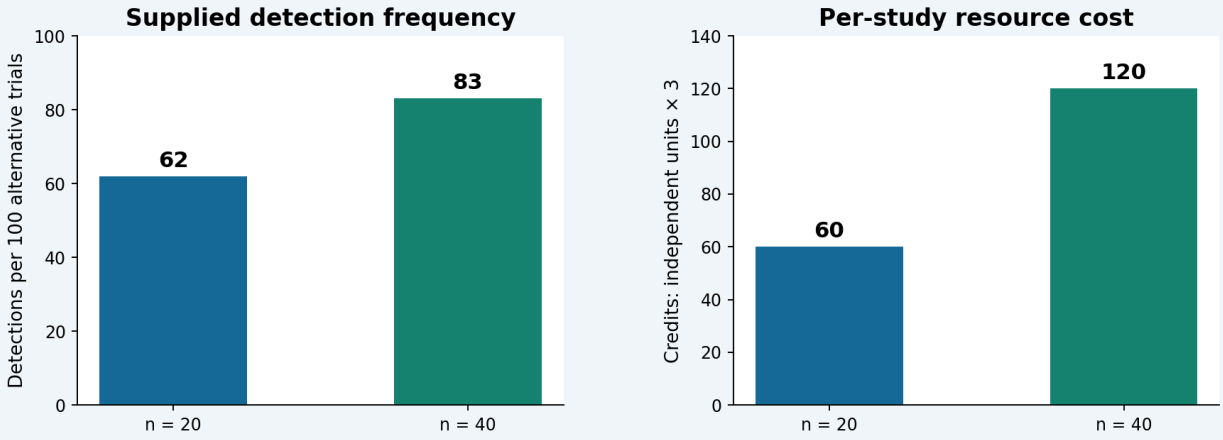
There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

The separate picture uses one hundred stipulated alternative trials per design, with sixty detections at n = 20 and eighty at n = 40. Its unit cost is two resource units, unlike the main example’s three credits. Estimated detection frequencies are 60% and 80%; per-study costs are forty and eighty resource units. The extra forty resource units accompany a twenty-percentage-point observed difference.

Neither is a measured effect on real pupils. Changing the assumed effect or variability may change power even at the same sample sizes, but no replacement detection counts are supplied. Null false-positive error needs separate null trials or appropriate theory.

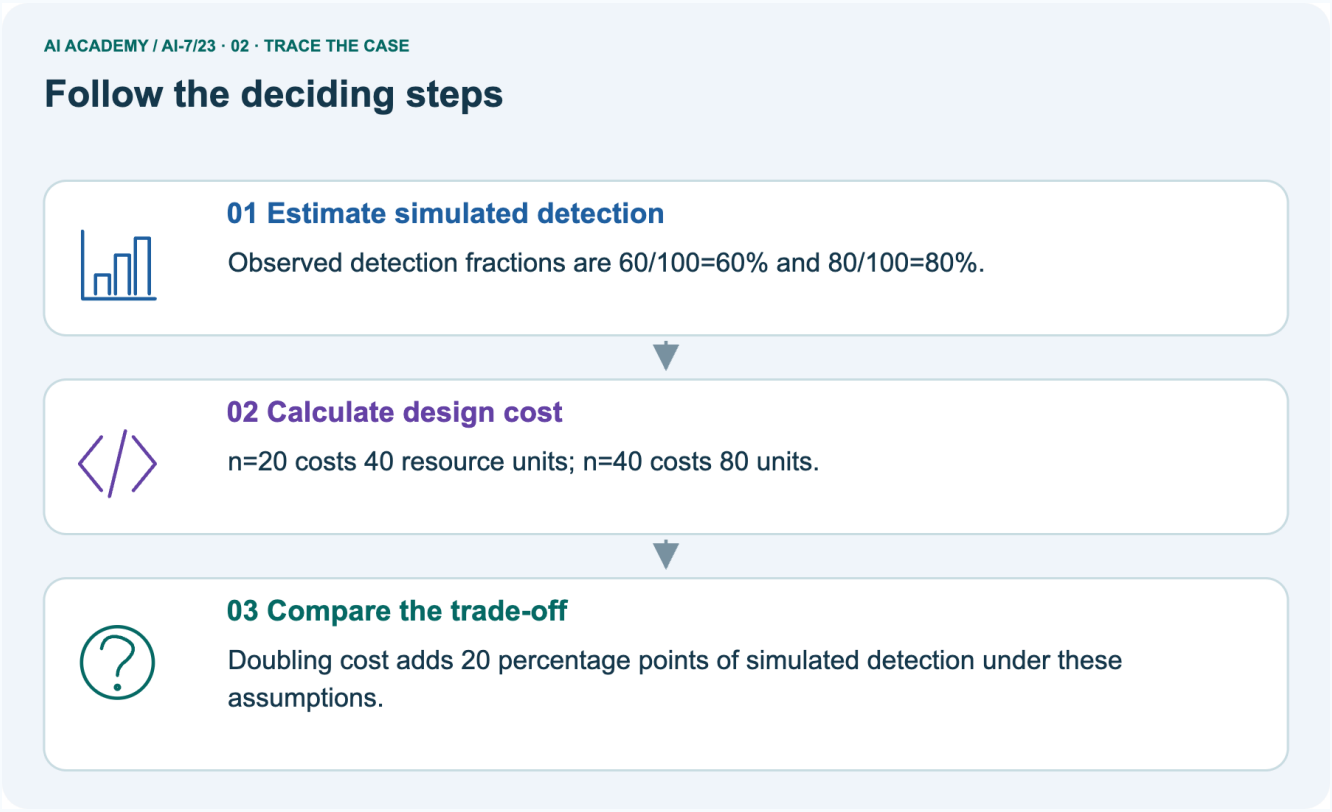
Separate picture quantity	n = 20	n = 40
Detections / supplied trials	60/100	80/100
Detection frequency	60%	80%
Cost per independent unit	2 resource units	2 resource units
Per-study cost	40 resource units	80 resource units
Large minus small	20 percentage points more detection	40 extra resource units
Changed effect/variability	New frequency unknown	New frequency unknown
What was actually run here	No simulations	No real study

Main example: detection estimates and cost are different quantities



Observed difference: 21 percentage points; extra per-study cost: 60 credits.
Counts are stipulated, not simulated here. Null error is unmeasured; real-study power is not guaranteed.

Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Estimate simulated detection	Observed detection fractions are 60/100=60% and 80/100=80%.
2. Calculate design cost	n=20 costs 40 resource units; n=40 costs 80 units.
3. Compare the trade-off	Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

Only 100 trials per design produce noisy estimates. They are fictional stipulated results, not executed simulation or **evidence** about pupils.

Change: Change the assumed effect or variability but keep the sample sizes.

Prediction: The detection fractions need new simulation; the resource costs stay 40 and 80.

Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Power	The test's detection probability under a specified alternative
Simulation trial	A generated dataset under a declared model
Planning effect	The effect size used for design calculations

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Power and resource tradeoffs

The facts we will use

AI ACADEMY / AI-7/23

Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/23

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are 60/100=60% and 80/100=80%.



2. Calculate design cost

n=20 costs 40 resource units; n=40 costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Power** and resource tradeoffs. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

The fractions are **60%** and **80%**; costs are 40 and 80 units. Forty extra units add 20 percentage points here. These simulation estimates do not establish real-study **power** without justified assumptions.

Changed-case reasoning: The detection fractions need new simulation; the resource costs stay 40 and 80. **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main: 62% versus 83% supplied alternative-trial detection frequencies, a twenty-one-percentage-point difference. Per-study costs are sixty and one hundred twenty credits, an extra sixty. Picture: 60% versus 80%, cost forty versus eighty resource units, twenty points for forty extra units. These are different supplied scenarios. Changed effect or variability needs new planning; no replacement frequency is known.

Independent task: 70/100 versus 85/100 gives 70% versus 85%, fifteen points; n = 30 or 60 at two per unit costs sixty or one hundred twenty. Null error is unmeasured, finite-trial estimates are noisy, and no real-study guarantee or universal sample-size claim follows.

Diagnose the current misconceptions

The worked result establishes every future case: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: One hundred alternative trials detect 70 at n 30 and85 at n 60; unit cost 2. Compare.

Teacher explanation: **70%** versus85%; cost 60 versus 120. Larger n improves detection here but not automatically under every model.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Power** concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

W2 Trace the worked case

For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

W3 Audit the limit

A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured. Qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

One hundred alternative trials detect 70 at n 30 and85 at n 60; unit cost 2. Compare.

Teacher explanation: **70%** versus85%; cost 60 versus 120. Larger n improves detection here but not automatically under every model.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: **70%** versus85%; cost 60 versus 120. Larger n improves detection here but not automatically under every model. Require intermediate reasoning and the stated scope. Fresh retry: Fifty alternative simulations detect35; sample n25, cost4 each. Expected reasoning: Estimated detection**70%**; study cost100 credits. Finite simulation uncertainty remains. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **power** and resource tradeoffs **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **power** and resource tradeoffs.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Reproducible environment and records. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/24, ai-7/26, ai-7/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **power** and resource tradeoffs

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Check every learner before discussion: Use the worked case for **Power** and resource tradeoffs. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the assumed effect or variability but keep the sample sizes. Require a **prediction** and a reason before comparing.

Expected result: The detection fractions need new simulation; the resource costs stay 40 and 80. Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

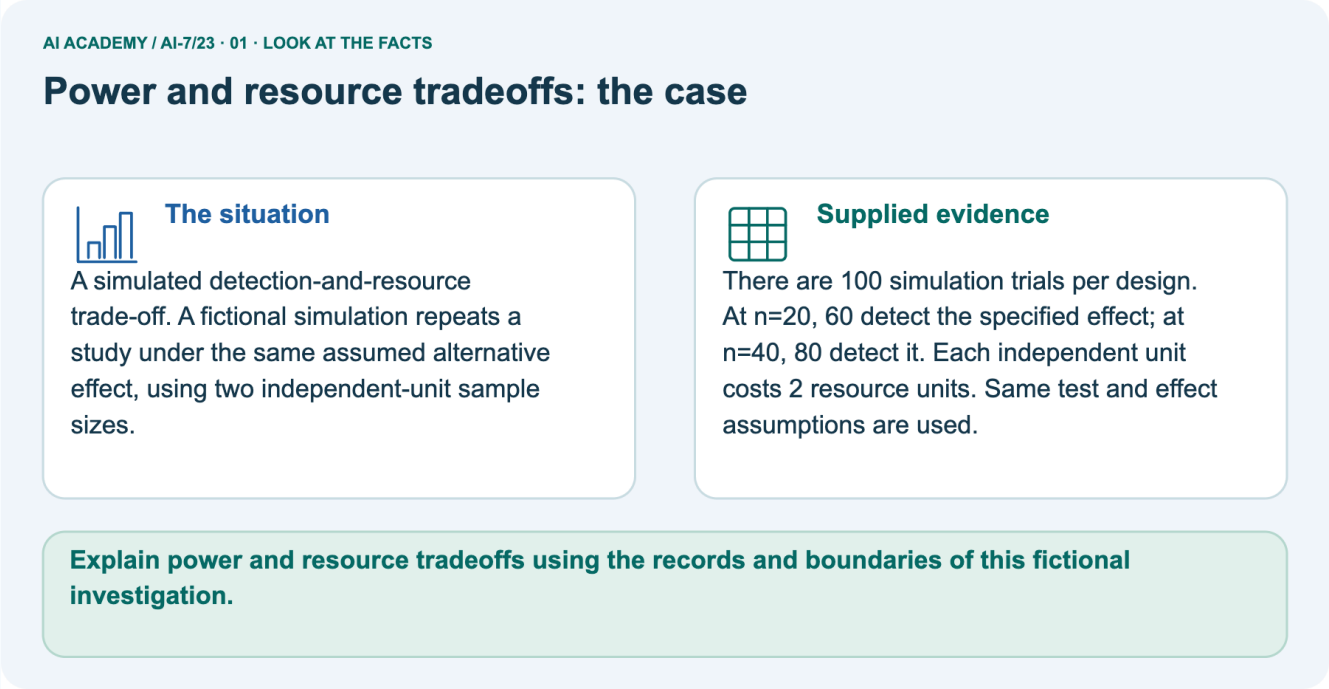
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Power and resource tradeoffs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

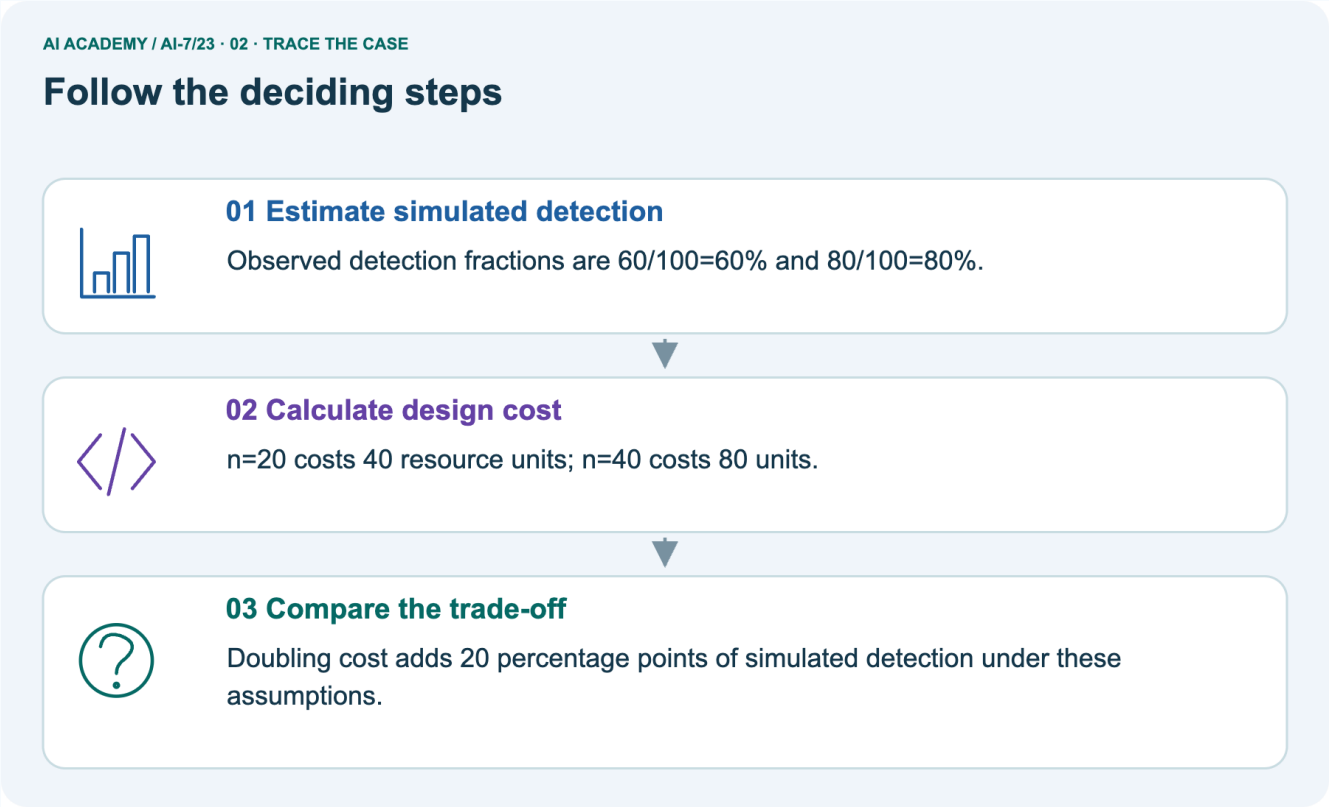
A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Supplied facts: There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



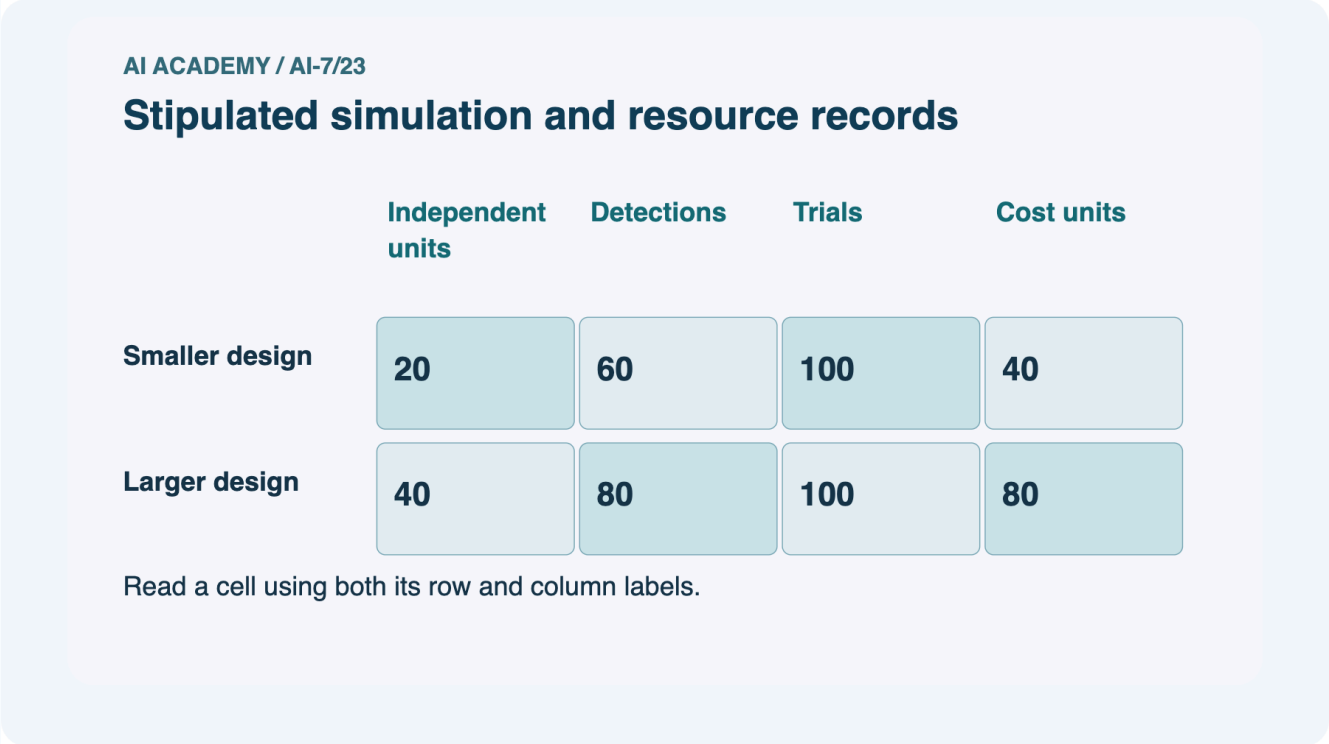
Follow the labelled picture with the matching explanation. Open the vector diagram

Estimate simulated detection: Observed detection fractions are 60/100=**60%** and 80/100=**80%**. → Calculate design cost: n=20 costs 40 resource units; n=40 costs 80 units. → Compare the trade-off: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Stipulated simulation and resource records



Follow the labelled picture with the matching explanation. Open the vector diagram

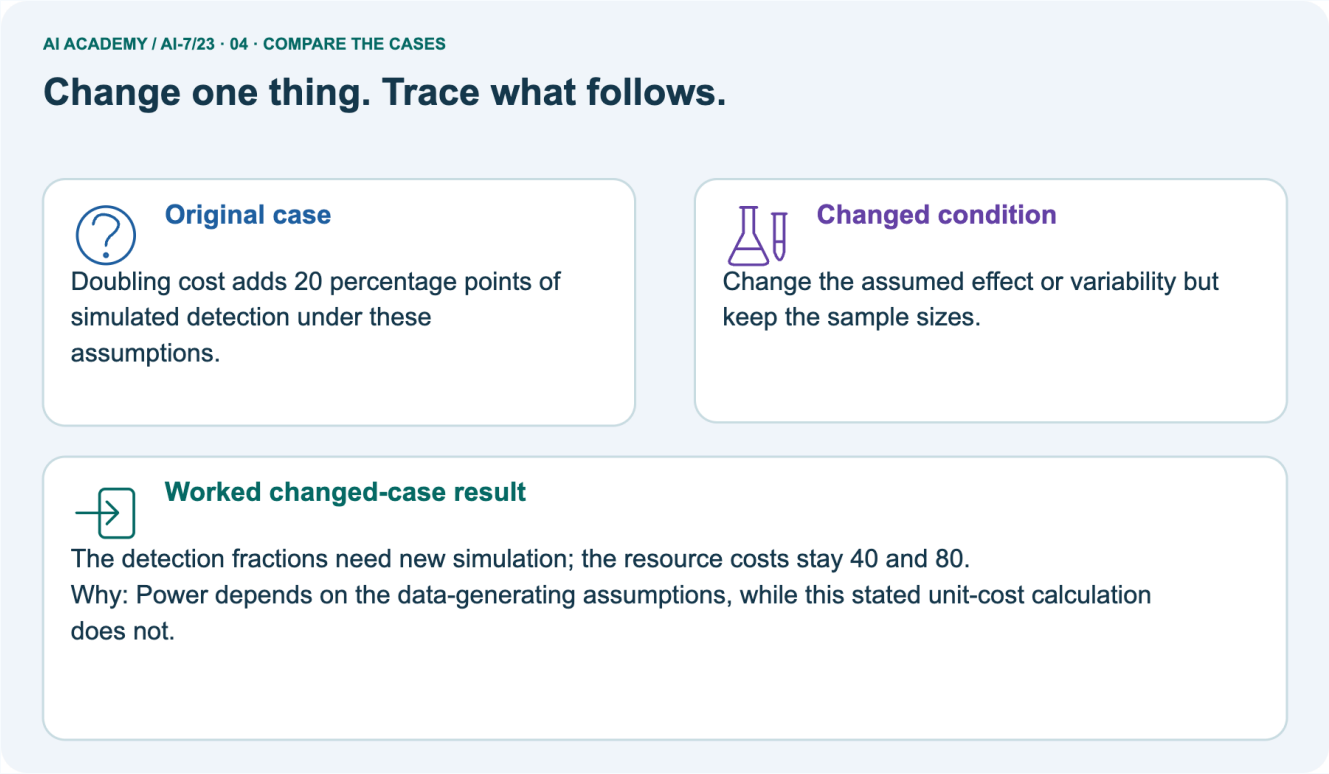
Stipulated simulation and resource records. Smaller design: Independent units = 20, Detections = 60, Trials = 100, Cost units = 40. Larger design: Independent units = 40, Detections = 80, Trials = 100, Cost units = 80.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Doubling cost adds 20 percentage points of simulated detection under these assumptions. Changed condition: Change the assumed effect or variability but keep the sample sizes.

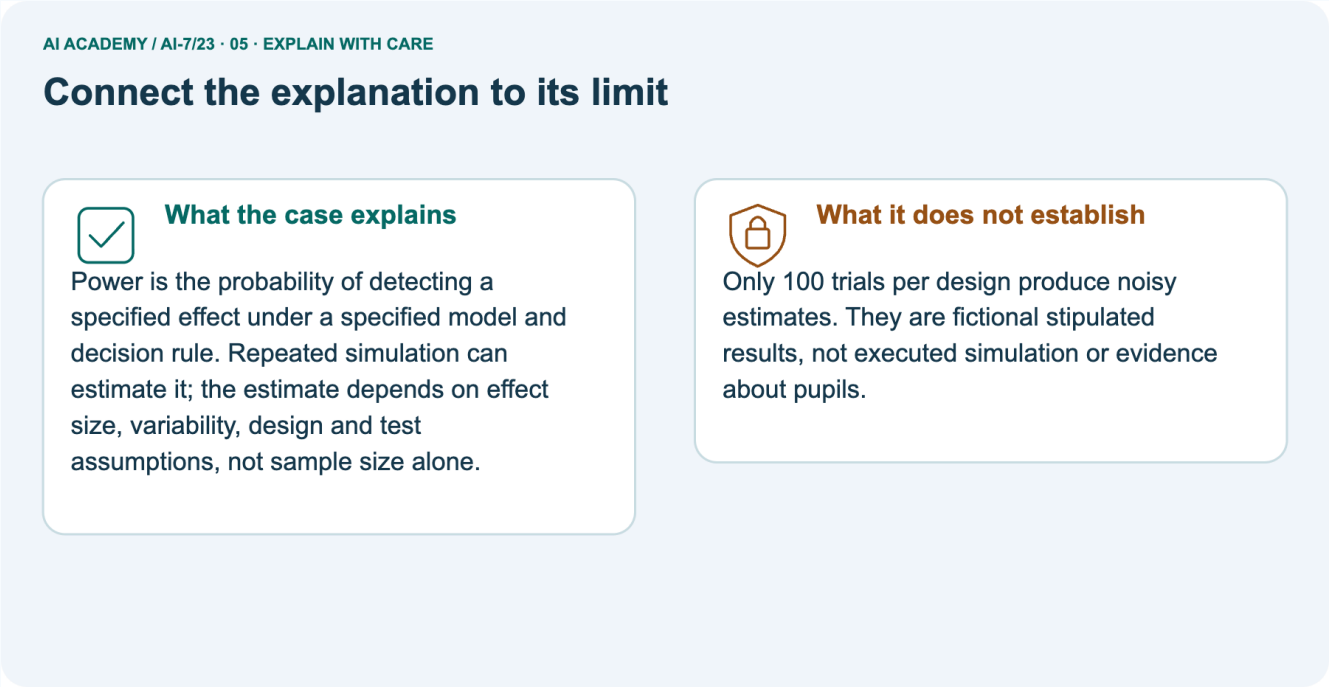
Expected result: The detection fractions need new simulation; the resource costs stay 40 and 80. Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone. Limit: Only 100 trials per design produce noisy estimates.

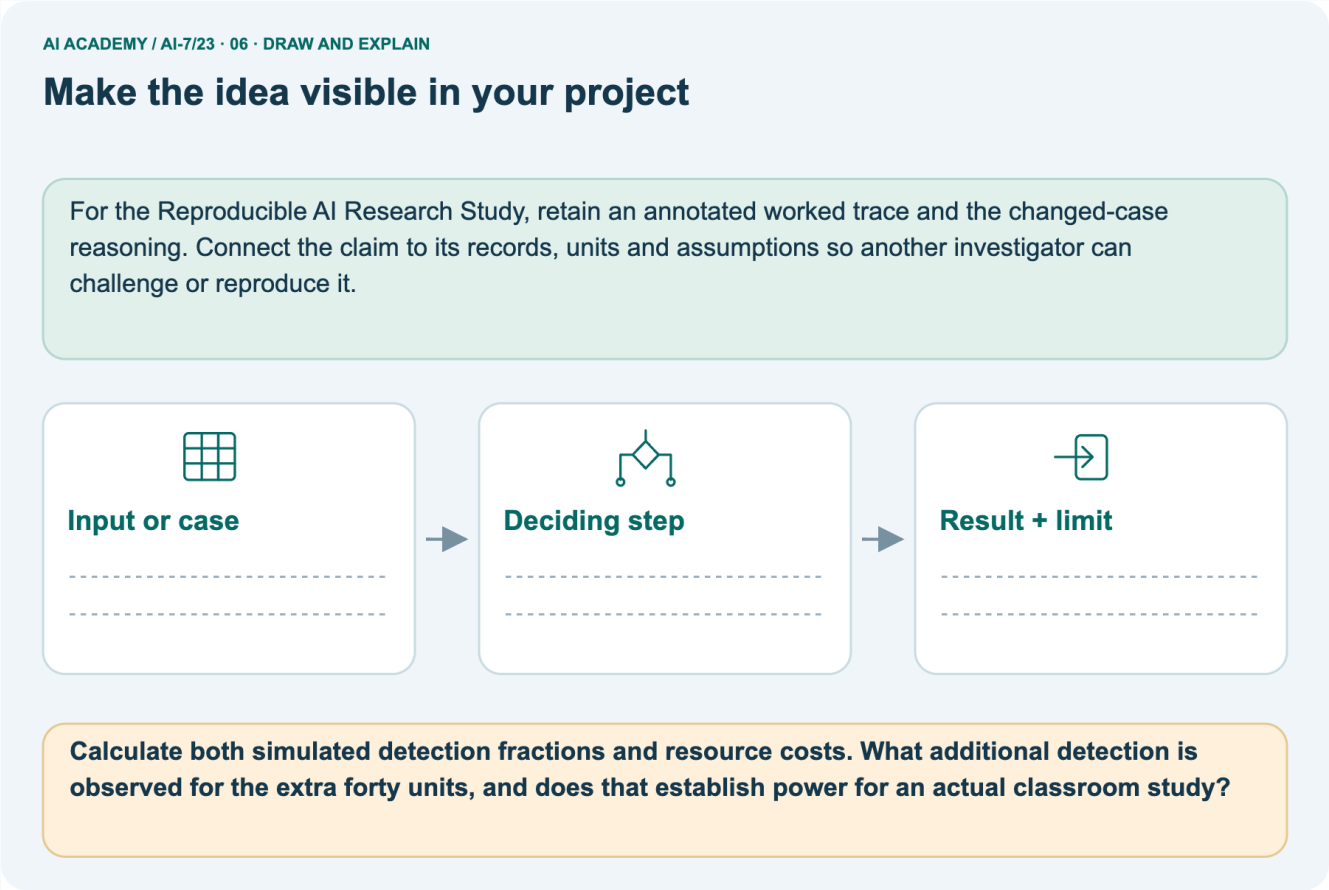
They are fictional stipulated results, not executed simulation or **evidence** about pupils.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Only 100 trials per design produce noisy estimates. They are fictional stipulated results, not executed simulation or **evidence** about pupils.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The fractions are **60%** and **80%**; costs are 40 and 80 units. Forty extra units add 20 percentage points here. These simulation estimates do not establish real-study **power** without justified assumptions.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Power and resource tradeoffs: the case

02. Follow the deciding steps

03. Stipulated simulation and resource records

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.